

VOLCANO QUEST

Explore, Predict & Survive

Unity Android Project Roadmap · 9 Weeks · May 19 – Jul 20, 2026

9
Levels

535
Max XP

9
Badges

2
3D Scenes

Free
Budget

PROJECT ROADMAP

■ Phase 1 – Setup & Foundation

Week 1–2 | May 19 – Jun 1

Setup

Install Unity & Android Build Tools

Unity 2022 LTS (free), Android SDK, JDK via Unity Hub. Enable Android Build Support module.

Setup

Create Project Structure

Scenes/, Scripts/, Assets/3D, Assets/UI, Assets/Audio, Prefabs/, Materials/

Config

Configure Android Settings

Package: com.yourname.volcanoquest | Min SDK: Android 8.0 | Target: Android 13+

UI

Global UI System

Main Menu, Level Select, HUD (XP bar, progress bar, badge display), scene transition manager.

Code

Game Manager Script

Persistent singleton: player XP, current level, unlocked levels, badge list. Saved via PlayerPrefs.

3D Asset

AI 3D Volcano Model

Meshy.ai (free) — prompt: 'cross-section volcano 3D, stylized, visible magma chamber'. Export .glb, import to Unity.

■ Phase 2 – Core Scenes (Flat UI)

Week 3–4 | Jun 2 – Jun 15

Level

Level 1 – Research Base

3 glowing tap objects. On tap: reveal objective + glow animation. Award 10 XP each. Unlock badge.

Level

Level 2 – Knowledge Scanner

5-item MCQ quiz. ScriptableObjects for questions. 15 XP per correct answer. Unlock badge on pass.

Level

Level 4 – Philippine Volcano Explorer

Static Philippines map (PNG). 5 volcano icon buttons. Popup facts + 1 question. 30 XP each.

Level

Level 6 – Volcano Builder

Match descriptions to volcano type (Shield, Cinder Cone, Composite). Step-by-step build animation. 50 XP.

Level

Level 7 – Volcano Investigation

4 volcano cards. Classify as Active/Inactive. Glowing monitor effect on correct. 60 XP.

Code

Reusable Quiz Prefab

One flexible MCQ prefab driven by ScriptableObjects. Reused for Levels 2, 9 and Assessment.

■ Phase 3 – 3D Interactive Levels

Week 5–6 | Jun 16 – Jun 29

Level	Level 3 – Lava Word Builder Lava river background (looping sprite). Letter tiles via IBeginDragHandler/IDragHandler/IDropHandler. Lava burst particle on correct word. 20 XP/word.
3D Core	Level 5 – Volcano Anatomy Lab ■ PRIMARY 3D LEVEL 3D cross-section volcano. Touch-rotate. Drag 6 label prefabs onto named anchor points. Correct: magma flow VFX + lock. Wrong: shake animation. 40 XP.
VFX	Volcano Particle System Unity Particle System: smoke (grey, rising), lava explosion (orange/red burst), ash cloud (dark spreading). Saved as reusable Prefabs.
Level	Level 8 – Eruption Warning Lab 4 image cards (shaking ground, red crater, smoke, bulging). Tap correct sign. Flashing alarm animation. Instant feedback. 70 XP.
Code	Drag & Drop System Generic DragDropManager.cs for Level 3 (letters→slots) and Level 5 (labels→3D anchors). Snap-back on wrong drop.

■ Phase 4 – Polish, XP & Badges

Week 7–8 | Jun 30 – Jul 13

System	XP & Progress System Progress bar animates on XP gain (DOTween). Totals: L1=30, L2=75, L3=80, L4=150, L5=40, L6=50, L7=60, L8=70, Assessment=80. Max=535 XP.
System	Badge System 9 badges: Volcano Explorer, Lava Investigator, Vocabulary Badge, Eruption Expert, Magma Analyst, Volcano Architect, Lava Bridge Champion, Eruption Warning Specialist, Volcano Master.
Feature	Avatar & Leaderboard Scientist avatar on HUD. Local leaderboard via PlayerPrefs. Display: avatar, level, badges, XP.
Audio	Audio Background volcano ambient. Correct chime. Wrong buzz. Eruption SFX. Free assets from freesound.org.
Level	Lesson Screen (Between L3 and L4) Scrollable text panel — volcano structure content. Interactive comprehension questions. 25 bonus XP.
Level	Level 9 – Final Assessment 5-question MCQ using reusable Quiz Prefab. 80 XP. Award Volcano Master Badge. Final score screen.

■ Phase 5 – Test & Export

Week 9 | Jul 14 – Jul 20

QA	Device Testing Test on physical Android via USB (Build & Run). Verify touch drag on L3 and L5. Check particle FPS.
QA	Performance Optimization Bake lighting on static scenes. ETC2 texture compression for Android. Disable shadows on mobile. Target 60 FPS.
Deploy	Build APK File > Build Settings > Android > Build. Signed APK or AAB for submission via Unity keystore manager.
Docs	Final Documentation Short user manual: APK install, game controls, level flow diagram. For academic submission.

GAME LEVEL FLOW

Level	Scene Name	Mechanic	XP	Badge
1	Research Base	Tap / Glow Objects	30	Volcano Explorer
2	Knowledge Scanner	MCQ Quiz	75	Lava Investigator
3	Lava Word Builder	Drag & Drop Letters	80	Vocabulary Badge
■	Lesson Screen	Read + Q&A;	+25	—
4	Philippine Explorer	Map Tap + Quiz	150	Eruption Expert
5	Volcano Anatomy ■3D	3D Label Drag	40	Magma Analyst
6	Volcano Builder	Type Matching	50	Volcano Architect
7	Investigation Center	Classify Volcanoes	60	Lava Bridge Champ
8	Eruption Warning Lab	Tap Warning Sign	70	Warning Specialist
9	Final Assessment	MCQ Quiz	80	■ Volcano Master

TOTAL POSSIBLE XP

535 XP

KEY NOTES



3D Levels

Level 3 (word drag) and Level 5 (anatomy lab) use 3D/advanced drag. All other levels are flat UI — keeps scope manageable.



AI 3D Model

Use Meshy.ai free tier — generate from text prompt, download .glb, import into Unity Assets/3D/ folder. No modelling skills needed.



Free Audio

Download royalty-free volcano ambient, chime, buzz, and explosion SFX from freesound.org. Import as AudioClip to Unity.



Particle VFX

Reusable Unity Particle System prefabs: lava burst, smoke cloud, ash spray. Triggered by C# events from drag-drop system.



Save System

PlayerPrefs stores XP, level unlock state, and badge flags. No backend or internet connection required for academic submission.



Tools Used

Unity 2022 LTS (free Personal), Android Studio SDK, Meshy.ai, freesound.org, DOTween (free asset), Unity Particle System.